

Task 11 Project Report

Basic Data Sorting & Filtering

1. Introduction

Task 11 focused on Basic Data Sorting & Filtering using Microsoft Excel. The purpose of this task was to practice basic data exploration by sorting and filtering retail sales data to answer simple business questions.

The task helped me understand how Excel filters can be used to analyze specific records without deleting or changing the original dataset.

2. Objective

The main objectives of this task were:

- * To understand basic data sorting.
- * To apply filters to sales data.
- * To use multiple filters together.
- * To answer simple business questions from the dataset.
- * To preserve the original/raw data unchanged.
- * To practice basic data exploration using Excel.

3. Tools Used

- * Microsoft Excel / Excel for Web
- * Retail Sales Dataset
- * Excel Table
- * Sorting and Filtering features

4. Dataset Description

The retail sales dataset contains information about customer orders, products, regions, quantities, sales, and profit.

The main columns used in the analysis were:

- * Order Date
- * Order ID
- * Region
- * State
- * Customer Name
- * Category
- * Sub-Category
- * Product Name
- * Quantity
- * Unit Price
- * Sales
- * Profit

5. Data Preparation

I opened the retail sales dataset in Excel and converted the data into an Excel Table. This enabled filter dropdowns for each column.

The original data was kept unchanged in the Original data sheet. A separate Filtered_data sheet was used to work with filtered records.

This approach helped preserve the raw dataset while allowing me to perform data exploration.

6. Sorting and Filtering

I used Excel's sorting and filtering features to explore the dataset.

The following activities were performed:

1. Sorted sales values to identify high-value sales records.
2. Applied a Region filter to view records from a specific region.
3. Applied multiple filters to narrow down the records.
4. Used Category, Sub-Category, Product Name, and other fields for exploration.
5. Compared sales and profit values to identify important records.

Multiple filters were useful for viewing only the records that matched specific conditions.

7. Business Questions

I prepared five business questions based on the retail sales dataset:

****Question 1:****

Which region generated the highest total sales?

****Question 2:****

Which product generated the highest total sales?

****Question 3:****

Which category sold the highest total quantity?

****Question 4:****

Which product generated the highest total profit?

****Question 5:****

Which region generated the highest total profit?

These questions can be answered using Excel filters, sorting, and PivotTables.

8. Observations from the Provided Data

From the data reviewed during the task:

- * The dataset contains sales records from multiple regions including Central, East, South, and West.
- * Technology products such as Smartphone, Business Phone, iPad, Gaming Laptop, Business Laptop, and tablets are present.
- * The dataset contains both Technology and Furniture categories.
- * The visible records include quantities ranging from 1 to 7 units.
- * Sales and profit values vary depending on the product and quantity sold.
- * Sorting and filtering made it easier to identify high-value sales and profit records.

9. Interview Questions

****Q1. When would you use filtering instead of deleting rows?**

I would use filtering when I want to temporarily view specific records without deleting the original rows. Filtering allows me to analyze selected data while keeping the complete dataset available.

****Q2. Why should raw data be preserved?****

Raw data should be preserved because it contains the original information. It can be used for validation, auditing, comparison, correction, and future analysis.

10. Key Learnings

Through this task, I learned:

- * How to sort data in ascending and descending order.
- * How to apply filters in Excel.
- * How to use multiple filters together.
- * How to analyze sales and profit information.
- * How to identify useful records from a large dataset.
- * Why raw data should not be deleted or unnecessarily modified.
- * How basic Excel features can support business data analysis.

11. Conclusion

Task 11 gave me practical experience with ****basic data sorting and filtering in Excel**. I learned how to explore retail sales data, apply multiple filters, sort important values, and use the results to answer business questions.

The task also helped me understand the importance of keeping raw data unchanged and performing analysis on a separate filtered view. This is an important practice in data analytics because it keeps the analysis organized, reproducible, and easy to verify.
